

Triphasic waves in the Harvard EEG reports — MedGemma-27B annotation

We ran the same triphasic-wave annotation we built for the Fraser Health reports over the three Harvard hospitals — MGH, BWH, and BCH, about 101,600 reports — with MedGemma-27B (Q4, grammar-constrained, temperature 0). Each report gets a status (present, explicitly absent, or not mentioned) and, when present, a phenotype (typical, atypical, mixed, or unspecified). As on the FHA set we ran it twice, with a short prompt and a longer one, to check the result does not depend on the wording, and we apply the same guard: a present or absent call has to have the triphasic term in the report text, otherwise it becomes not mentioned.

About 98–99% of reports got a valid annotation in both prompts; the ~1–2% gap at MGH and BWH is almost all very long continuous-monitoring reports that run past the model’s context window (BCH had none). The numbers below are the guarded short-prompt labels.

How often triphasic waves appear

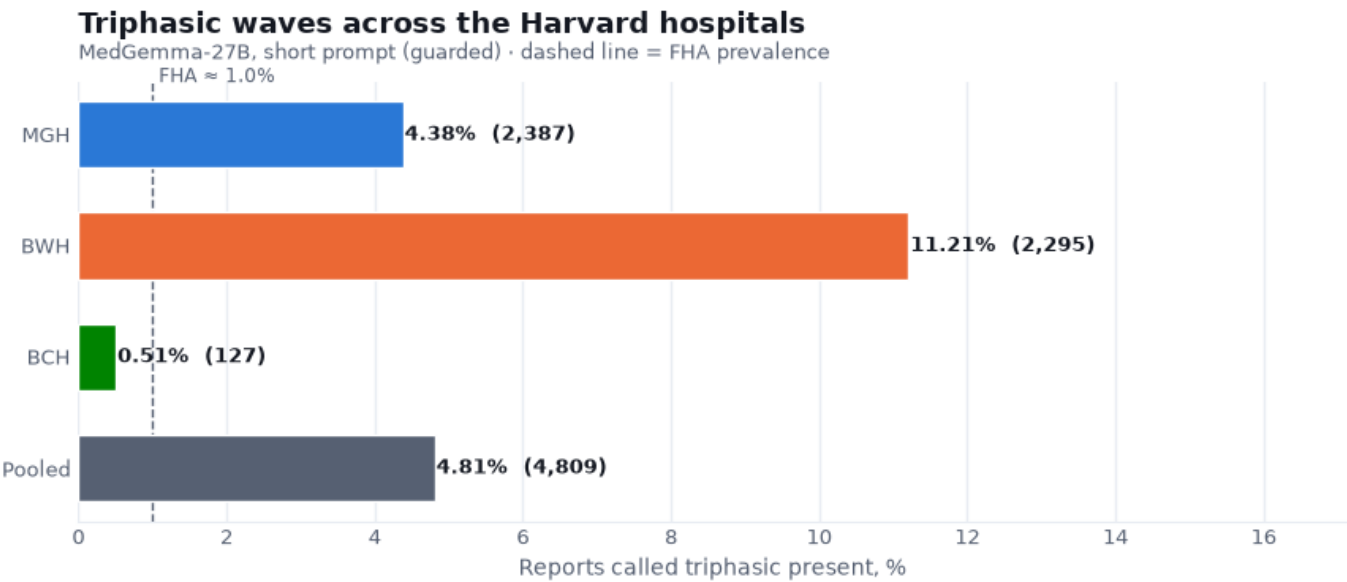


Figure 1: Prevalence by hospital

Hospital	Reports	Present	Present %	Explicitly absent	Not mentioned
MGH	54,452	2,387	4.38%	26	52,430
BWH	20,474	2,295	11.21%	29	18,372
BCH	25,022	127	0.51%	7	24,888
Pooled	99,948	4,809	4.81%	62	95,690

The rate varies a lot by hospital, and it is much higher than the ~1% we saw at Fraser Health. BWH stands out at 11%, MGH at about 4%, while the pediatric hospital BCH is only 0.5%. This tracks the patients: triphasic waves are a marker of metabolic and anoxic encephalopathy, common in the critical-care and inpatient populations of large adult academic centers and rare in children. So the difference is about who is being recorded, not about the model behaving differently — the reports at BWH and MGH simply describe triphasic waves far more often.

What kind of triphasic waves

Pooled across the three hospitals, of the reports called present the phenotype is **unspecified 4,115, typical 404, atypical 256, mixed 34**. As on the FHA set, most reports that mention triphasic waves do not describe enough morphology to sub-type them, so the great majority land in unspecified.

Do the two prompts agree



Figure 2: Prompt stability

Hospital	Exact status agreement	present/absent κ
MGH	99.93%	0.99
BWH	99.85%	0.99
BCH	99.95%	0.95
Pooled	99.92%	0.99

The two prompts give the same status on more than 99.8% of reports everywhere, with κ around 0.99 (0.95 at BCH, where present cases are few). This is the same result we got on the FHA set, now on a different hospital system: the annotation barely moves with the wording of the prompt.

Faithfulness

The guard flipped **367 reports** (MGH 267, BWH 91, BCH 9) from present or absent to not mentioned — cases where the model inferred triphasic waves from an anoxic or metabolic context with no triphasic wording in the text. That is the same error mode we saw on the FHA reports, and the same one-line text check removes it; after the guard every present or absent label has the term in the report.

In short

Triphasic waves are documented in about 5% of the Harvard reports pooled, but that hides a wide spread — 11% at BWH, 4% at MGH, 0.5% at the pediatric hospital — driven by the patient population, not the model. The annotation is stable to the prompt (99.9% agreement, $\kappa = 0.99$), the same as on Fraser Health, and the one recurring error, reading triphasic waves into an anoxic or metabolic picture, is small and filtered out. There is no human ground truth here, so these are the model's reading of the reports.

Reproduce

```
python -m analysis.triphasic_harvard_analysis      # the tables above and the two figures
```

Labels are produced by `slurm/submit_triphasic_harvard.sh` (via `slurm/label_triphasic.sbatch`), which runs `src/cpu/triphasic.py` reading the report text from the HEEDB zips through the cohort indexes. Output: `results/triphasic_harvard/<COHORT>/<VARIANT>/.`

Pipeline, prompts, grammar, and the scripts behind this report: **<https://github.com/vbronetskyi/eeg-report-classification>**